

Towards Collaborative Machine Learning in Far-Edge and Edge Environments

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Abstract. In a world where AI is increasingly integrated into everyday applications, the strategy for developing and deploying state of the art AI models has been relying heavily on powerful cloud-based servers. However, this approach has several drawbacks, including the distance between the data source, the server and the end-user, which can lead to increased latency, higher energy consumption, and potential privacy concerns. To address these issues, edge computing has emerged as a promising solution, bringing computational resources closer to where they are needed, in particular, the TinyML paradigm, which provides a possibility to run AI models on everyday devices. Nonetheless, far-edge devices, such as microcontrollers, although highly memory and computationally constrained, are widely used in the IoT ecosystem, equipped with sensors and actuators, and are capable of running small tasks while consuming very low power. Faced with such devices, the possibility of running a DNN seems infeasible, but with the right strategies, it is possible to leverage the capabilities of these devices and orchestrate them to achieve this task. This work explores the potential of distributed inference, where a DNN model is partitioned and distributed across multiple resource-constrained devices to achieve highly performant inference in the edge environment. We present DisRUPT, a distributed inference testbed, that enables the evaluation of different partitioning strategies and deployment scenarios for DNN models incorporating far-edge and edge devices together, in regards of inference latency and network energy consumption. Additionally, we present a set of strategies applied on different DNN architectures, and evaluate their effects on inference latency and the network energy consumption. In particular, we managed to run an adapted MobileNetV1 model on a network of 2 microcontrollers and 1 Raspberry Pi 3B, executing an inference in 760,32 ms (33,78 ms of active inference and 726,54 ms of overhead latency), using only 145,7 mJ of energy. From our setups with the DS-CNN model we obtained results that ranged from low latency, taking 27,99 ms to low energy consumption inference, using 26,5 mJ of energy per inference. From the collected results in this work, we identify a set of best practices for distributed inference in far-edge and edge devices, and, mainly, point out the existence of an input throughput threshold where the inclusion of edge devices in the inference process only becomes advantageous above the threshold.

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ACM Computing Classification System:

- Computer systems organization → Embedded and cyber-physical systems

- Computer systems organization → Distributed architectures
- Computing methodologies → Neural networks